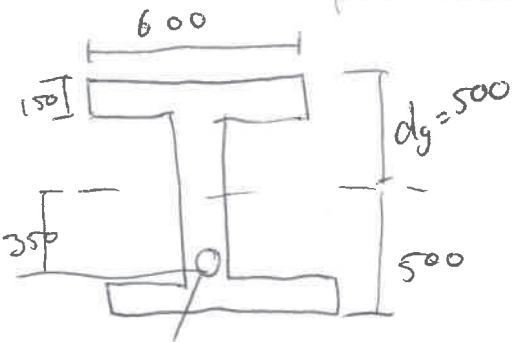
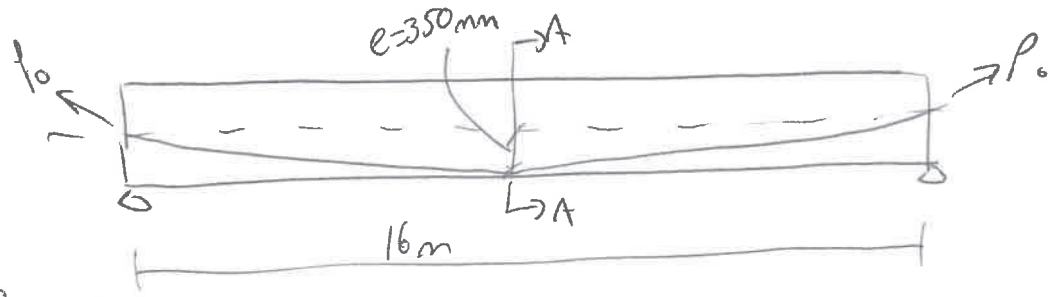


Problem 6 - Calculation of Prestress losses.



10 x 12.7mm super grade low relaxation strand.

$$P_j = P_0 = 10 \times 184 \times \frac{70}{100} = 1288 \text{ kN}$$

P_0 (jacking force) $\xrightarrow{\text{immediate losses}}$ P_i (initial prestress force) $\xrightarrow{\text{deferred losses}}$ P_e (effective prestress force)

Short-term (Immediate) losses

$\Rightarrow$ Elastic losses: since beam is post-tensioned with only one cable, no elastic losses

$\Rightarrow$ Friction loss

$$\sigma_{pa} = \sigma_{pj} e^{-\mu(\alpha_{tot} + \beta_p \cdot L_{pa})} \quad | \quad \text{cl 3.4.2.4}$$

$$\sigma_{pj} = \text{jacking stress in tendon} = \frac{P_j}{A_p} = \frac{1288 \times 10^3}{1000 \text{ mm}^2} = 1288 \text{ MPa}$$

μ = duct friction = 0.2 (zinc-coated flat metal duct)

β_p = wobble effect duct = 0.016 rad/m

α_{tot} = the total angular change over the length of cable

$$\alpha_{tot} = \frac{4h}{L} = \frac{4 \times 350}{16,000} = 0.0875 \text{ rad}$$

h = cable eccentricity

$$\sigma_{pa} = 1288 e^{-0.2(0.0875 + 0.016 \times 8)}$$

$$\sigma_{pa} = 1236 \text{ MPa} \quad \text{or} \quad \sigma_{pa} = 0.96 \sigma_{pj}$$

$$\% \text{ loss} = \left(\frac{1288 - 1236}{1288} \right) \times 100\% = 4\%$$

$\Rightarrow$ anchorage slip

6mm draw-in at the anchorage (from manufacturers information)

rate of ~~stress~~ loss, mf.

$$\text{loss in first half of beam} = 0.04 \times 1288 = 52 \text{ MPa}$$

$$mf = \frac{52 \text{ MPa}}{8 \text{ m}} = 6.4 \text{ MPa/m} \quad (0.0064 \text{ MPa/mm})$$

wedge-slip length

$$L_{ws} = \sqrt{\frac{E_p \cdot \delta_{ws}}{mf}} = \sqrt{\frac{190,000 \times 6}{0.0064}} = 13.3 \times 10^3 \text{ mm}$$

Stress loss at the anchorage due to wedge slip

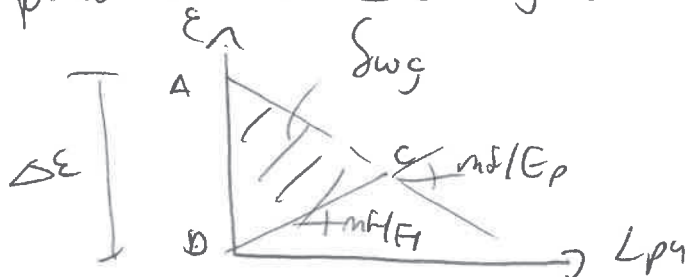
$$\Delta \sigma = 2mf \cdot L_{ws} = 2 \times 6.4 \times 13.3 = \underline{170 \text{ MPa}}$$

This reduces along the tendon length because of friction and at mid-span ($L_{pa} = 8 \text{ m}$)

section loss is approximately

$$170 \times \frac{(13.3 - 8)}{13.3} = 65 \text{ MPa}$$

which is about 5% $\left(\frac{65}{1288} \times \frac{100}{1} \right)$ of the prestress force at jacking.



Deferred losses

⇒ concrete shrinkage.

loss of prestress due to concrete shrinkage.

$$\Delta \sigma_p = E_{cs} \frac{E_p}{\alpha} \quad (\text{cl 3.4-3.2})$$

where α is the modification to allow for reinforcement effect.

$$\alpha = 1 + 15 \frac{A_s}{A_g} = 1 + 15 \times \frac{3600}{320,000} = 1.17$$

E_{cs} = Design shrinkage strain (cl 3.1.7.1)

$$= E_{cse} + E_{csd}$$

$$e_h = \text{hypothetical thickness} = \frac{2A}{u_e} = \frac{2 \times 320,000}{(600 \times 2 + 50 \times 4 + 700 \times 2 + 400 \times 2)} = 160 \text{ mm}$$

$$- \underline{E_{cse}} = \text{autogenous shrinkage strain (cl 3.1.7.2(1))} \\ = E_{cse}^* \times (1.0 - e^{-0.01t})$$

$$t = 30 \text{ years} \times 365 \text{ days} = 10,950 \text{ days}$$

E_{cse}^* - final autogenous shrinkage strain

$$= (0.07 f'_c - 1.0) \times 50 \times 10^{-6} \quad (\text{cl 3.1.7.2(3)})$$

$$= (0.07 \times 32 - 1.0) \times 50 \times 10^{-6} \\ = 62 \times 10^{-6}$$

$$E_{cse} = 62 \times 10^{-6} \times (1.0 - e^{-0.1 \times 10,950}) \\ = 62 \times 10^{-6}$$

- ϵ_{csd} = drying shrinkage strain
 $= K_1 K_4 \epsilon_{csd.b}$ c1 3.1-7.2(7)

$K_1 = 1.32$ $t_h = 160 \text{ mm}$ (fig 3.1-7-2)

$K_4 = 0.5$ (near coastal area)

$\epsilon_{csd.b}^*$ = basic drying shrinkage strain
 $= 800 \times 10^{-6}$

$\epsilon_{csd} = 1.32 \times 0.5 \times 800 \times 10^{-6} = 528 \times 10^{-6}$

$\epsilon_{csd.b} = (0.9 - 0.005 t_h) \times \epsilon_{csd.b}^* = 592 \times 10^{-6}$

- $\epsilon_{cs} = \epsilon_{cse} + \epsilon_{csd} = (62 + 528) \times 10^{-6}$

$\epsilon_{cs} = 590 \times 10^{-6}$ 453×10^{-6}

$\Rightarrow$ loss of prestress due to shrinkage:

$\Delta \sigma_p = \epsilon_{cs} \frac{E_p}{\alpha} = 590 \times 10^{-6} \times \frac{190,000}{1.17}$

$= 94 \text{ MPa}$

% loss of prestress due to concrete

shrinkage $= \frac{94}{1288} \times 100\% = 7.3\%$

⇒ Loss of prestress due to concrete creep.

$$\Delta \sigma_p = \epsilon_{cc} E_p \quad (\text{cl 3.4.3.3})$$

$$\epsilon_{cc} = 0.8 \phi_{cc} \left(\sigma_{ci} / E_c \right) = \text{creep strain in concrete}$$

σ_{ci} = sustained stress in concrete at the level of prestress

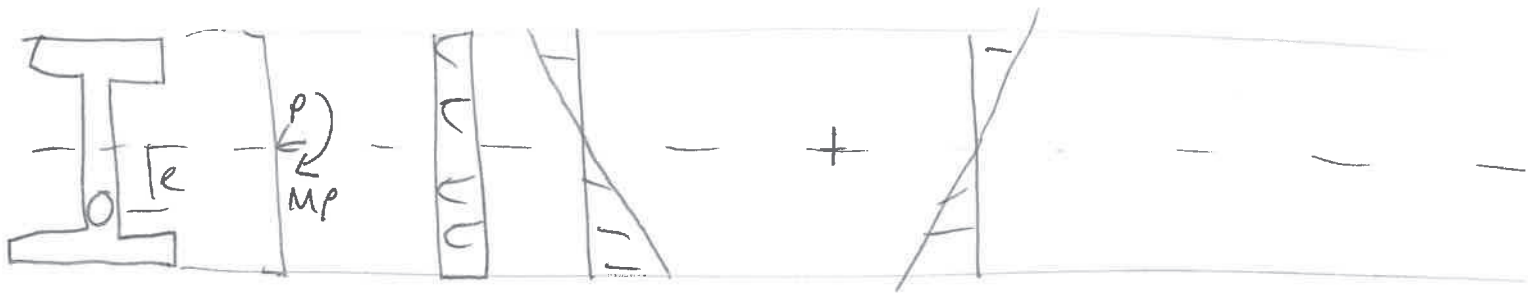
Assume only sustained load is self-weight of beam section

$$w_{sw} = 24 \text{ kN/m}^3 \times 320,000 \times 10^{-6} \text{ m}^2 = 7.68 \text{ kN/m}$$

$$M_{sw} = \frac{w l^2}{8} = \frac{7.68 \times 16^2}{8} = 246 \text{ kNm}$$

at mid-span

The stress in concrete at level of prestress due to prestress and sustained load



$$\sigma_{(y=e)} = \underbrace{\frac{P_i}{A_g} + \frac{P_i \cdot e \cdot e}{I_g}}_{\text{prestressing load only}} - \underbrace{\frac{M_{sw} \cdot e}{I_g}}_{\text{sustained load}} = \sigma_{ci}$$

P_i = initial prestress force

$$= P_j \times \% \text{ immediate losses (9\%)}$$

$$= 1288 \times \left(1 - 9/100 \right) = 1172 \text{ kN}$$

$$\sigma_{ci} = \frac{1172 \times 10^3}{320,000} + \frac{1172 \times 10^3 \times 350 \times 350}{3.86 \times 10^{10}} - \frac{246 \times 10^6 \times 350}{3.86 \times 10^{10}}$$

$$\sigma_{ci} = 3.7 + 3.72 - 2.23$$

$$\sigma_{ci} = 5.2 \text{ MPa (compression)}$$

$\Rightarrow \phi_{cc} = \text{Design creep co-efficient.}$

$$= K_2 K_3 K_4 K_5 K_6 \phi_{cc.b} \quad (\text{cl 3.1.8.3})$$

$$\phi_{cc.b} = 3.4 \quad (f_{lc} = 32) \quad (\text{tb 3.1.8.2})$$

$$t = 30 \text{ years}$$

$$t_h = 160 \text{ mm}$$

$$K_2 = 1.29$$

(Fig 3.1.8.3)

$$K_3 = \frac{2.7}{[1 + \log(t)]} = 1.103$$

where $t = 28 \text{ days}$ when sustained load first applied.

$$K_4 = 0.5$$

$$K_5 = 1.0$$

$$K_6 = 1.0$$

$$\therefore \phi_{cc.b} = 1.29 \times 1.103 \times 0.5 \times 1 \times 1 \times 3.4 = 2.4$$

$\Rightarrow$ the creep strain

$$\epsilon_{cc} = 0.8 \phi_{cc} \left(\frac{\sigma_{ci}}{E_c} \right) = 0.8 \times 2.4 \times \left(\frac{5.2}{30,100} \right) = 0.00033$$

Hence the loss of prestress due to creep of concrete

$$\Delta \sigma_p = \epsilon_{cc} \cdot E_p = 0.00033 \times 190,000 = 63 \text{ MPa}$$

$$\% \text{ loss of prestress due to concrete creep} = \frac{63}{1288} \times 100\% = 4.9\% \quad \textcircled{d}$$

⇒ Tendon stress relaxation

For low relaxation strand. take

$$R_b = 3\%$$

Design relaxation of a tendon R is

$$R = K_7 K_8 K_9 R_b \quad (\text{cl 3.3.4.3})$$

$$K_7 = \log(5.4(5)^{1/6}) = \log(5.4(10950)^{1/6}) = 1.4$$

$$K_8 = 0.5 \quad \left(\text{for } \frac{\sigma_{pi}}{f_{pb}} = \frac{1172}{1870} \approx 0.6 \right) \quad (\text{tb 3.3.1})$$

$$f_{pb} = 1870 \text{ MPa}$$

$$K_9 = 1.0$$

$$\therefore R = 1.4 \times 0.5 \times 1.0 \times 3 = 2.1\%$$

$$\text{loss of prestress due to tendon relaxation} = R \left(1 - \frac{\text{loss of stress due to creep \& shrinkage}}{\sigma_{pi}} \right)$$

$$= 2.1 \left(1 - \frac{74 + 63}{1172} \right) = 1.83\%$$

⇒ Summary of losses

$$\begin{aligned} \text{friction loss} &= 4\% \\ \text{anchorage slip} &= 5\% \end{aligned}$$

(short-term immediate losses)

$$\text{shrinkage} = 4.0\%$$

$$\text{creep} = 4.9\%$$

$$\text{Relaxation} = 1.83\%$$

(deferred losses)